

backtrack.

Find the missing skill.

Get back to today's lesson.

THE PROBLEM

A student gets stuck, but doesn't know which earlier skill to review.

Check today's
task

Return to today's lesson
with a fresh check

Practice the missing skill
with Khan Academy

The search can become the study session

01 WHAT THEY NEED TO FINISH

Solve this equation

$$x^2 + 7x + 12 = 0$$

They search "quadratics".



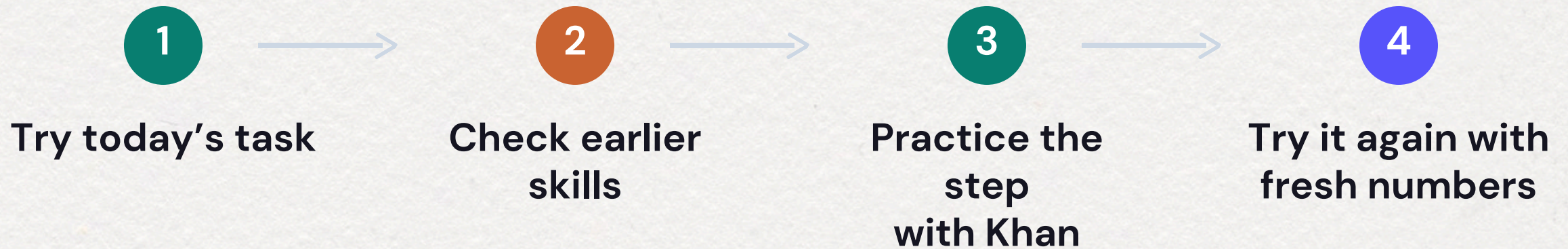
02 WHAT MAY BE BLOCKING IT

Find the factor pair

Multiply to 12. Add to 7.

BACKTRACK checks the gap before choosing the repair.

How BACKTRACK gets a learner unstuck



Answers choose the next step. Fresh problems check the return.

Two wrong answers. Different help.

Tried $x = 3, 4$



Check signs → fresh goal

Tried $x = 2, 6$



Check factors



Fresh goal

The mistake points to the starting check.

Khan material, focused on the missing step



1

Watch the key part

1:34 focused segment
Match sum and product



2

Try the step here

Use the idea with
different numbers



3

Return to the goal

Solve a fresh quadratic
equation

Short enough to use now. More Khan practice when they need it.

Back to the skill that stopped them

FRESH CURRENT-LEVEL TASK

$$x^2 + 11x + 28 = 0$$

$$x = -4 \text{ or } x = -7$$

New numbers. The same skill they came to recover.

Try the demo. Watch the route change.

Compare two wrong answers.
See why they get different repairs.

Try BACKTRACK

backtrack-learning.vercel.app

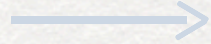
A few sample routes to help you picture BACKTRACK. The full product is still in development.



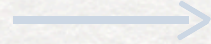
Scan to start

Can they still do it next week?

1



2



3

**Stuck before the
route**

**Pass a fresh task
after practice**

**Pass again 7–14
days later**

Retained reentry

Count everyone who started unable to do the task.

A five-month school pilot

80–120 learners across 2–3 cohorts



November 2026–March 2027

Pilot targets, fitted to each school's access and timetable.

Does the route add value?

Khan practice

Selected by the teacher

Khan + BACKTRACK

Practice tied to the recovery route

```
graph TD; A[Khan practice] --- B[ ]; B --- C[The same fresh assessments]; D[Khan + BACKTRACK] --- B;
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The diagram illustrates two parallel practice routes. On the left, a light green box labeled 'Khan practice' with the subtext 'Selected by the teacher' is connected by a green line to a central dark green box labeled 'The same fresh assessments'. On the right, a light purple box labeled 'Khan + BACKTRACK' with the subtext 'Practice tied to the recovery route' is connected by a purple line to the same central box. The lines from both boxes meet at a horizontal line, which then drops vertically to the central box.

The same fresh assessments

After practice, then 7–14 days later. Similar time and support.

Skip the review you don't need.

What the learner does

What changes

Know the step



Skip its review

Repair a gap



Move closer to the goal

Come back later



Keep earlier progress

Progress changes what you need to do next.

One routine. Many learning goals.

Fractions

Algebra

Graphs

Every new destination uses a reusable kit

A reviewed map + fresh checks + matched Khan material

Another educator can run the same recovery routine.

For classes, self-study, and the next topic a learner needs.

A budget built around learners

PHP 100,000

60%

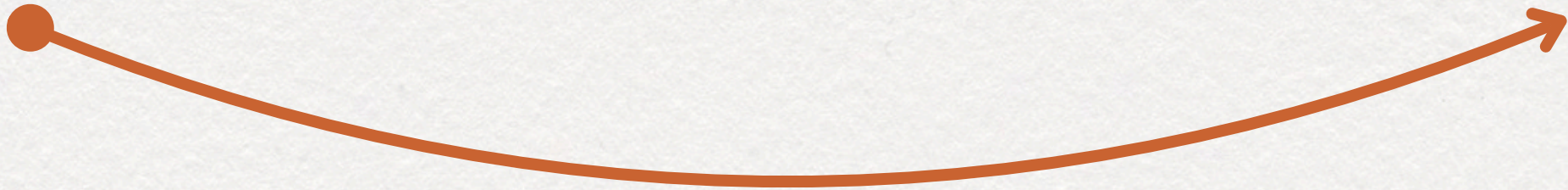
for learning review, access and evaluation

Learner access stays free.

Allocation	PHP
Learning review	24,000
Access support	18,000
Evaluation	18,000
Deployment	16,000
Product operations	8,000
Teacher onboarding	8,000
Recruitment content	3,000
Contingency	5,000
Total	100,000

backtrack.

A route back to today's lesson.



University of the Philippines Manila

Matthew Labrador, Paul Recio, Harry Gomez

Coach: Justin Mesias

Research behind the approach

Khan Academy teacher reports

Spacing and retrieval · 2022

ALEKS custom objectives

Confidence assessment · Foster

Gemini study notebooks · 2026

Self-Determination Theory in education

IES learning practice guide